



You are an expert clinical causal inference researcher, medical ML scientist, and causal representation learning specialist.

Your task is to construct the most clinically plausible latent-variable causal DAG for an irregular medical time-series dataset.

Dataset name:
MIMIC-III

Official dataset description:
DATASET-DESCRIPTION-LINK = <https://physionet.org/content/mimiciii/1.4/>

GLOBAL RESEARCH MISSION — READ BEFORE DOING ANYTHING

Your final goal is NOT merely to construct rule-based decision trees.

Your final goal is to construct a research-backed latent-variable causal model for MIMIC-III.

The decision trees are only an implementation mechanism for converting observed measurements into binary proxy labels for latent clinical states. They are not the scientific endpoint.

The scientific endpoint is:

1. A clinically plausible latent causal ontology:
 - What are the true hidden clinical states behind the observed measurements?
 - Which latent states are meaningful causal variables?
 - Which observed variables are only noisy manifestations of these states?
2. A clinically and causally justified latent DAG:
 - How do background variables affect latent clinical states?
 - How do latent states affect each other?
 - How do latent states affect mortality?
 - How do latent states generate observed labs, vitals, and charted values?
 - How does missingness/measurement intensity arise from latent severity and clinical decision-making?
3. A causal-inference-ready graph:
 - Which variables are confounders?
 - Which variables are mediators?
 - Which variables are colliders?
 - Which variables should or should not be adjusted for when estimating effects on mortality?
 - Which latent states are outcome-proximal severity markers rather than valid adjustment variables?
4. A practical operationalization layer:
 - Only after the latent variables are selected and justified, create rule-based binary decision trees.
 - These trees produce implementable proxy labels such as LAT_SHOCK, LAT_RENAL_DYSFUNCTION, and LAT_GLOBAL_SEVERITY.
 - The trees are tools for implementation, not the main research contribution.

Priority order:

1. Research-backed latent clinical states.
2. Research-backed latent causal DAG.
3. Measurement model from latent states to observed variables.
4. Missingness and irregular-sampling model.
5. Confounding/mediation/collider analysis.
6. Rule-based decision trees for implementation.

If there is any conflict between making a simple decision tree and preserving causal correctness, preserve causal correctness.

Core objective:

Construct a clinically grounded latent causal representation of the dataset, including:

1. A set of causal latent variables representing underlying patient clinical states.
2. Formal definitions of these latent variables using available observed measurements and metadata.
3. Rule-based binary decision trees that convert observational/meta data into binary latent-variable labels.
4. A directed acyclic graph (DAG) over background variables, latent clinical states, observed measurements, measurement/missingness processes, and mortality.
5. A clear explanation of how the latent clinical states causally affect mortality.
6. A clear explanation of how the observed variables are generated as manifestations, measurements, or proxies of the latent clinical states.

Important conceptual rule:

The observed variables in the dataset are not the main causal variables. They are clinical measurements, test results, charted values, or proxies. The causal variables of interest are latent clinical states such as shock, respiratory failure, renal dysfunction, hepatic dysfunction, inflammation/sepsis burden, coagulation dysfunction, metabolic derangement, neurologic dysfunction, cardiac injury, global severity, chronic baseline risk, or other dataset-appropriate latent states.

Do not construct a naïve causal graph where every lab/vital directly causes every other lab/vital. Instead, construct a causal representation in which:

- Background/static variables influence latent patient states.
- Latent patient states influence observed measurements.
- Latent patient states influence mortality.
- Observed measurements are mainly manifestations of latent physiology, not primary causal mechanisms.
- Measurement frequency and missingness may themselves be informative and influenced by latent severity, clinical decision-making, or care intensity.

Your goal is to approximate the true underlying clinical causal structure as well as possible using deep research, medical reasoning, causal inference principles, and the official dataset documentation. Do not merely produce a correlation graph.

PHASE 1 — Dataset audit

First, open and study DATASET-DESCRIPTION-LINK carefully.

Extract:

1. The complete list of available variables.
2. Which variables are static/background variables.
3. Which variables are time-varying observed measurements.
4. Which variables are outcomes, especially mortality.
5. Which variables are interventions or care-process variables, if any.
6. Units, clinical meaning, valid ranges, and measurement windows when available.
7. Known dataset collection rules and missingness patterns.
8. Whether the dataset contains ICU type, admission type, care unit, demographics, diagnoses, chart events, labs, vitals, medications, procedures, or outcomes.

Then summarize the dataset in a compact audit table.

Do not proceed to latent-variable construction until the dataset audit is complete.

PHASE 2 — Deep Research protocol and research goals

Perform deep research before constructing latent variables, decision trees, or the DAG.

The purpose of Deep Research is not to collect thresholds for decision trees.

The purpose of Deep Research is to infer the most plausible latent clinical causal structure behind the dataset.

Your research must answer the following research questions:

RQ1 — Latent-state discovery:

What hidden physiological or clinical states plausibly generate the observed measurements in MIMIC-III?

Examples:

- Shock or hemodynamic instability may generate abnormal blood pressure, heart rate, lactate, urine output, and vasopressor-related signals if available.
- Respiratory failure may generate abnormal oxygenation, respiratory rate, blood gas, ventilation, and saturation signals.
- Renal dysfunction may generate abnormal creatinine, BUN, urine output, potassium, bicarbonate, and acid-base patterns.
- Inflammation/sepsis burden may generate fever/hypothermia, WBC abnormalities, lactate elevation, shock, and organ dysfunction.
- Global severity may represent multi-system physiological instability rather than a single measurement.

RQ2 — Measurement model:

For each observed variable, determine whether it is:

- a background/static cause,
- an observed proxy of a latent state,
- a treatment/intervention/care-process variable,
- a measurement/missingness-process variable,
- an outcome,
- or irrelevant/noisy for the causal model.

Do not treat observed labs and vitals as the primary causal variables unless they are true interventions.

RQ3 — Latent causal DAG:

Determine the causal relationships among:

- background variables,
- latent clinical states,
- mortality,
- observed measurements,
- and missingness/measurement processes.

The graph should represent a clinical data-generating process:

background risk → latent physiology → observed measurements and mortality.

RQ4 — Mortality mechanism:

Explain how mortality is caused through latent clinical states, not through raw measurements directly.

For example:

- chronic baseline risk → vulnerability → global severity → organ dysfunction → mortality
- acute insult → inflammation/sepsis → shock → renal/metabolic failure → mortality
- respiratory failure → hypoxemia/acidosis → global severity → mortality

Adapt these mechanisms to the actual dataset variables.

RQ5 — Missingness and irregular sampling:

Because MIMIC-III is an irregular clinical time-series dataset, research whether missingness is likely:

- MCAR,
- MAR,
- MNAR,
- or caused by care intensity / clinician suspicion / latent severity.

Treat measurement frequency and missingness as potentially informative, not as random noise.

RQ6 — Operational proxy construction:

Only after RQ1-RQ5 are answered, construct binary rule-based decision trees for the approved latent variables.

The decision trees should approximate the latent states using available observed variables, but they must not replace the causal DAG.

Research sources to use:

1. DATASET-DESCRIPTION-LINK.
2. Dataset-specific papers, benchmark papers, and preprocessing conventions.
3. ICU severity literature: SOFA, APACHE, SAPS, sepsis, shock, organ failure, respiratory failure, renal failure, coagulation, hepatic dysfunction, neurologic dysfunction, metabolic derangement.
4. Causal inference literature: SCMs, DAGs, latent variables, backdoor adjustment, mediation, colliders, potential outcomes.
5. Causal representation learning literature: observed variables as functions/proxies of latent causal variables.
6. Clinical EHR/time-series literature: irregular sampling, informative missingness, measurement intensity, care processes.
7. Existing clinical causal graphs or phenotype networks, if available.

For every proposed latent variable, decision rule, and DAG edge, provide:

- clinical mechanism,
- dataset support,
- causal direction,
- whether it risks leakage,
- whether it is a confounder, mediator, collider, or measurement proxy,
- confidence level,
- and whether it should be kept, optional, or rejected.

Do not proceed from research to final DAG until you can explain why the selected latent variables are the right causal level for this dataset.

PHASE 3 — Core causal ontology

Use the following node types:

A. Background/static variables

Examples:

- Age
- Sex/gender
- Ethnicity/race, if available
- Height, weight, BMI
- ICU type, admission type, first care unit
- Chronic disease burden or baseline vulnerability, if derivable

Background variables may cause latent clinical states and mortality risk, but they are generally not caused by ICU measurements.

B. Latent clinical state variables

These are the main causal variables.

Possible candidate latent states include, but are not limited to:

- Chronic baseline risk / chronic burden
- Acute insult / acute physiological stress
- Global severity of illness
- Circulatory shock / hemodynamic instability
- Respiratory failure / pulmonary dysfunction
- Renal dysfunction / kidney injury
- Hepatic dysfunction / liver injury
- Coagulation or hematologic dysfunction
- Inflammation / infection / sepsis burden
- Neurologic dysfunction
- Cardiac injury / cardiac strain
- Metabolic derangement / acid-base-electrolyte disorder
- Nutritional, fluid balance, or volume-status burden, if supported by variables
- Treatment/care intensity, if the dataset contains care-process variables

Do not force this list. Adapt the latent variables to the actual variables in MIMIC-III.

Each latent variable must have:

1. A clinical definition.
2. A causal interpretation.
3. Observable indicators from the dataset.
4. Suggested proxy rule or thresholding logic.
5. Expected direction of abnormality.
6. Confidence level.
7. Notes on possible ambiguity.

C. Observed measurement variables

These include labs, vitals, charted measurements, and test results.

Treat these as manifestations or noisy measurements of latent states, for example:

- Shock may manifest through low blood pressure, high lactate, tachycardia, vasopressor use if available.
- Respiratory failure may manifest through PaO₂, FiO₂, oxygen saturation, respiratory rate, ventilation variables if available.
- Renal dysfunction may manifest through creatinine, BUN, urine output, potassium, bicarbonate.
- Hepatic dysfunction may manifest through bilirubin, AST, ALT, albumin.
- Coagulation dysfunction may manifest through platelets, INR/PT/PTT if available.
- Inflammation/infection may manifest through WBC, temperature, lactate, cultures/antibiotics if available.
- Cardiac injury may manifest through troponin, heart rate, blood pressure, ECG-related variables if available.
- Metabolic derangement may manifest through glucose, bicarbonate, pH, electrolytes, anion gap if derivable.

An observed variable may be an indicator of more than one latent state. Allow multi-parent measurement edges when clinically justified.

D. Mortality outcome

Mortality should be treated as an outcome node.

Latent states may cause mortality directly.

Background variables may affect mortality directly or indirectly through latent states.

Observed measurements should generally not directly cause mortality unless the variable is truly an intervention or treatment exposure. Labs and vitals usually indicate the severity state rather than causing death themselves.

E. Measurement and missingness process

Because MIMIC-III is an irregular medical dataset, include a measurement process if relevant:

- Measurement intensity
- Test ordering process
- Missingness indicator
- Observation availability
- Care intensity

These may be influenced by latent severity, clinician suspicion, ICU type, care unit, admission type, or background risk.

Do not assume missingness is MCAR.

Discuss whether missingness is likely MAR or MNAR.

If missingness is clinically informative, include it as a separate observation-process node, not as a biological causal state.

PHASE 4 — Latent variable selection and definition

Before constructing the DAG, define the latent variables that will appear in the DAG.

This phase must happen before DAG construction.

Your task in this phase:

1. Propose candidate latent clinical variables based on the dataset audit.
2. Map available observed and metadata variables to candidate latent variables.
3. Remove candidate latent variables that are not sufficiently supported by the dataset.
4. Merge overlapping latent variables if the dataset cannot distinguish them reliably.

5. Mark weakly measured but clinically important latent variables as optional.

6. Finalize the latent variable set before constructing causal edges.

For every latent variable, define it operationally from observed data.

For each latent variable produce:

1. Name
2. Clinical meaning
3. Why it is latent rather than directly observed
4. Observable indicators
5. Metadata indicators, if relevant
6. Suggested binary definition
7. Suggested continuous severity score, if useful
8. Thresholds or quantile-based fallback rules
9. Missingness handling
10. Directionality: higher value means worse/better?
11. Whether this latent variable should be used as:
 - prediction target,
 - confounder,
 - mediator,
 - outcome-proximal severity marker,
 - treatment-effect modifier.
12. Confidence: high / medium / low.
13. Whether to keep, merge, make optional, or reject.

Definitions must be clinically meaningful but also implementable on MIMIC-III.

Do not define latent variables from a single measurement unless unavoidable.

Prefer multiple indicators and majority-vote or score-based definitions.

If the dataset lacks sufficient indicators for a latent state, either:

- remove the latent variable,
- mark it as weakly measured,
- merge it with another latent variable,
- or define it as optional.

Do not use mortality or post-outcome information to define any latent variable.

Do not allow outcome leakage.

Do not use future measurements if the latent label is intended to represent an earlier clinical state.

At the end of this phase, provide a final approved latent-variable list.

Only these approved latent variables may be used as latent nodes in the DAG.

IMPORTANT — Decision trees are not the final goal

The rule-based binary decision trees are only an operational layer.

They are used to create binary proxy labels for latent variables so that the thesis pipeline can generate `latent_tags.csv` and use those labels in downstream modeling.

Do not optimize the entire answer around the decision trees.

A correct answer with clinically meaningful latent variables and a strong DAG is better than a large collection of simplistic decision trees.

The decision trees must serve the causal model, not define the causal model.

Before writing any tree, explicitly state:

- which latent clinical state the tree approximates,
- why this state belongs in the DAG,
- which observed variables are manifestations of that state,
- and how the proxy label may differ from the true unobserved clinical state.

PHASE 5 — Rule-based binary decision trees for latent labels

After the latent-variable set is finalized, construct a rule-based binary decision tree for each approved latent variable.

Use the uploaded image `decision-tree-example.png` as the structural template.

The desired structure is:

1. Define several clinically meaningful domain indicators.
2. Each domain indicator is a Boolean condition based on observed measurements or metadata.
3. Combine domain indicators into a score:
`score = sum(domain indicators)`
4. Apply a threshold:
if `score >= THRESHOLD`:
 `LATENT_VARIABLE = 1`
else:
 `LATENT_VARIABLE = 0`

Example structure from `decision-tree-example.png`:

- Circulatory domain:
 `MAP_first < 70 OR SysABP_first <= 100`
- Neurologic domain:
 `GCS_first < 15`
- Respiratory domain:
 `RespRate_first >= 22 OR SaO2_first < 92`
- Metabolic domain:
 `Lactate_first > 2 OR pH_first < 7.3 OR HCO3_first < 18`
- Renal domain:
 `Creatinine_first >= 2`
- `score = sum(domain indicators)`
- `score >= 2` gives Severity = 1
- otherwise Severity = 0

For each latent variable, create a similar binary decision tree.

Requirements:

1. The output of every tree must be binary: 0 or 1.
2. The rules must use only observed variables and metadata available in MIMIC-III.
3. Use clinically interpretable thresholds when possible.
4. If standard clinical thresholds are not available or not dataset-appropriate, propose quantile-based fallback thresholds.
5. Each tree should use multiple indicators when possible.
6. A tree may use first value, worst value, mean value, slope, variability, measurement frequency, or missingness indicators, but you must justify each choice.
7. Avoid using mortality, survival time, discharge disposition, or outcome-derived variables.
8. Avoid using future values if the latent variable is meant to represent early-state physiology.
9. Use missingness explicitly when clinically informative, but do not let missingness dominate the latent definition unless justified.
10. Mark rules as high / medium / low confidence.

For each decision tree, provide:

1. Latent variable name.
2. Clinical interpretation of label 1.
3. Input variables.
4. Domain indicators.
5. Scoring function.
6. Binary threshold.
7. Missingness policy.

8. Pseudocode.
9. A compact Mermaid flowchart.
10. Clinical rationale.
11. Potential false positives and false negatives.
12. Confidence level.

The decision trees should be designed so they can be directly implemented in code to generate a latent_tags.csv file.

Use naming convention:

LAT_<LATENT_NAME> = 0/1

For example:

LAT_SHOCK = 1 means the patient has evidence of circulatory shock or clinically meaningful hemodynamic instability.

LAT_RENAL_DYSFUNCTION = 1 means the patient has evidence of acute or clinically meaningful renal dysfunction.

LAT_GLOBAL_SEVERITY = 1 means the patient has evidence of multi-domain severity according to the score-based tree.

After creating the decision trees, verify:

- Every approved latent variable has a binary tree.
- Every tree uses only allowed variables.
- No tree uses mortality or future outcome information.
- Every threshold is clinically or statistically justified.
- Every tree can be implemented on irregular time-series data.

Only after this verification, proceed to DAG construction.

PHASE 6 — DAG construction rules

Construct the DAG only after:

1. The dataset variables were audited.
2. The latent variables were selected and finalized.
3. The binary decision trees were constructed and validated.

The final graph must be a DAG.

Rules:

1. No directed cycles.
2. No edge from mortality back to clinical states.
3. No edge from future measurements to earlier latent states.
4. No automatic observed-variable-to-observed-variable causal chains unless there is a true clinical mechanism.
5. Prefer latent-state-mediated explanations over direct lab-to-lab causation.
6. Use temporal and clinical ordering:
 - Background risk precedes acute illness.
 - Acute insult precedes or contributes to global severity.
 - Global severity may contribute to organ dysfunction.
 - Organ dysfunction contributes to mortality.
 - Organ dysfunction produces observed abnormal measurements.
7. Allow clinically justified latent-to-latent edges, such as:
 - Chronic risk → acute vulnerability
 - Acute insult → global severity
 - Global severity → shock
 - Shock → renal dysfunction
 - Shock → hepatic dysfunction
 - Shock → metabolic derangement
 - Inflammation/sepsis → shock

- Inflammation/sepsis → coagulation dysfunction
 - Respiratory failure → metabolic derangement
 - Renal dysfunction → metabolic derangement
 - Cardiac injury → shock
8. Avoid over-dense DAGs. Use sparse, clinically meaningful edges.
 9. Each edge must have a mechanism and confidence score.
 10. If an edge is plausible but uncertain, mark it as optional or low-confidence.
 11. Do not confuse measurement with intervention.
 12. Do not confuse association with causation.
 13. Do not use the mortality label to define latent variables in a way that leaks the outcome.
 14. Observed variables should usually be children of latent states, not parents of latent states.
 15. Binary latent labels generated by the decision trees are operational proxies for latent states, not the biological states themselves. Make this distinction explicit.

DAG construction order:

1. Add background/static nodes.
2. Add approved latent clinical-state nodes.
3. Add mortality outcome node.
4. Add observed measurement nodes as manifestations of latent states.
5. Add measurement/missingness process nodes if needed.
6. Add background-to-latent edges.
7. Add latent-to-latent edges.
8. Add latent-to-mortality edges.
9. Add latent-to-observed measurement edges.
10. Add latent/background-to-measurement-process edges.
11. Check acyclicity.
12. Check clinical plausibility.
13. Check whether the graph is sparse enough.

PHASE 7 — Iterative validation procedure

Do not produce the final answer in one pass.

Use the following internal process:

Round 1:

Create an initial list of candidate latent variables from the official dataset variables.

Round 2:

Map every observed variable to one or more latent clinical states. Identify unassigned variables and decide whether they are irrelevant, background, outcome, treatment, or measurement-process variables.

Round 3:

Finalize the latent-variable set. Remove, merge, or mark optional latent variables that are poorly supported.

Round 4:

Construct rule-based binary decision trees for every approved latent variable using observed and metadata variables.

Round 5:

Validate the decision trees:

- Do the rules use only available variables?
- Are thresholds clinically meaningful?
- Is missingness handled correctly?
- Is there outcome leakage?
- Are the trees implementable on irregular time-series data?
- Are false positives/false negatives acceptable?

Round 6:

Construct a first-pass latent DAG using only approved latent variables. Check for cycles, clinically implausible edges, and over-dense structure.

Round 7:

Research again. Compare the graph to known clinical severity frameworks and EHR causal/phenotype-network literature. Revise latent variables, decision rules, and edges if needed.

Round 8:

Perform adversarial critique:

- Which latent variables are not truly identifiable from the dataset?
- Which decision-tree rules are too weak or too dependent on a single measurement?
- Which edges are only correlations?
- Which observed variables were incorrectly treated as causes?
- Which latent variables are poorly measured?
- Which missingness patterns may bias definitions?
- Which edges create collider bias if adjusted for?
- Which variables should not be adjusted for when estimating effects on mortality?
- Which variables are mediators rather than confounders?
- Which latent definitions risk outcome leakage?

Only after these rounds, produce the final answer.

PHASE 8 — Required final output

Your final answer must contain the following sections.

1. Dataset audit

A table with:

- Variable name
- Variable type: background / observed time-varying / outcome / treatment / measurement-process
- Clinical meaning
- Unit if available
- Notes

2. Candidate latent variables

A table with:

- Candidate latent variable name
- Clinical meaning
- Supporting observed variables
- Dataset support level
- Keep / merge / optional / reject
- Rationale

3. Final approved latent variables

A table with:

- Latent variable name
- Clinical meaning
- Observed indicators
- Suggested definition
- Missingness strategy
- Confidence: high / medium / low
- Rationale

4. Rule-based binary decision trees

For every approved latent variable, provide:

- Latent variable name
- Binary label definition
- Input variables

- Domain indicators
- Score formula
- Binary threshold
- Missingness policy
- Pseudocode
- Mermaid flowchart
- Clinical rationale
- False positive risks
- False negative risks
- Confidence level

The structure should resemble decision-tree-example.png:

domain indicators → score = sum(domain indicators) → threshold → binary label.

5. Observed-to-latent measurement model

A table with:

- Observed variable
- Parent latent variable or variables
- Direction of abnormality
- Whether it is specific or non-specific
- Whether it is used in a binary decision tree
- Notes

6. Latent causal DAG edge list

A table with:

- Source node
- Target node
- Edge type: background-to-latent / latent-to-latent / latent-to-observed / latent-to-mortality / measurement-process
- Clinical mechanism
- Confidence
- Source or reasoning basis
- Keep / optional / reject

7. Mortality mechanism summary

Explain the main causal paths to mortality, for example:

- Chronic baseline risk → global severity → organ dysfunction → mortality
- Acute insult → shock → renal/metabolic failure → mortality
- Inflammation/sepsis → shock/coagulation dysfunction → mortality
- Respiratory failure → hypoxemia/acidosis → mortality

Adapt these paths to the actual dataset and final DAG.

8. Missingness and irregular sampling analysis

Discuss:

- Which variables are likely MNAR or MAR.
- Whether missingness should be represented as a separate node.
- Whether measurement frequency can act as a proxy for severity or care intensity.
- How missingness should be handled when defining latent variables.
- How missingness enters or does not enter the binary decision trees.

9. Backdoor/confounding implications

For the purpose of estimating causal effects on mortality:

- Identify likely confounders.
- Identify likely mediators.
- Identify colliders to avoid conditioning on.
- Explain which latent variables should be used for adjustment and which should not, depending on the treatment/exposure of interest.

- Explain whether the binary latent tags are safe adjustment variables or whether they may be descendants/mediators.

10. Graph specification

Provide the final DAG as:

- A. Adjacency list.
- B. Mermaid graph.
- C. Python NetworkX edge list.

Use clear node prefixes:

- BG_ for background variables
- LAT_ for latent variables
- OBS_ for observed measurements
- MISS_ for missingness/measurement-process variables
- OUT_ for mortality outcome

11. Decision-tree implementation specification

Provide a code-oriented specification for generating latent_tags.csv:

- Required input columns.
- Derived summary features, such as first, worst, mean, slope, count, missingness.
- One function per latent variable.
- Expected binary output column names.
- Handling of missing variables.
- Unit assumptions.
- Threshold table.
- Validation checks.

12. Rejected alternatives

List important variables, latent definitions, decision rules, or edges that were considered but rejected, with reasons.

13. Final confidence and limitations

State:

- Which latent variables are strongest.
- Which are weakest.
- Which decision trees are most reliable.
- Which decision trees are most uncertain.
- Which edges are most uncertain.
- What additional data would improve identifiability.
- Why the graph is a clinically plausible causal DAG rather than a proven ground-truth DAG.

PHASE 9 — Style and rigor requirements

Be rigorous and skeptical.

Every final latent variable must be selected before the DAG is constructed.

Every final latent variable must have a binary decision tree.

Every final decision tree must be implementable in code.

Every final edge must be justified.

Every latent variable must be clinically interpretable.

Every observed variable must be categorized.

Do not hallucinate variables not present in the dataset.

Do not overclaim certainty.

Do not present the graph as statistically proven from observational data alone.

Do not ignore informative missingness.

Do not treat labs/vitals as primary causal states.

Do not create a dense graph just because variables correlate.

Do not define latent variables after creating the DAG.

Do not construct the DAG before the latent variables and binary decision trees are stable.

The final output should be detailed enough to implement in code:

- latent tag construction,

- rule-based binary decision trees,
- latent_tags.csv generation,
- DAG construction,
- confounder selection,
- mortality-effect estimation,
- sensitivity analysis.

Begin now by opening <https://physionet.org/content/mimiciii/1.4/> and auditing the dataset variables.

The clinically correct causal level for MIMIC-III is a sparse set of latent physiologic states—not a dense graph over raw labs and vitals—because the official database is a 26-table ICU EHR assembled from routine care and contains demographics, bedside vitals, laboratory results, procedures, medications, notes, imaging reports, and mortality rather than protocolized mechanistic variables. The strongest MIMIC-III latent DAG for mortality is therefore: background risk and case-mix → latent acute physiology and organ failure → observed measurements and care intensity → mortality, with an explicit measurement-process node because tests are ordered and charted during routine clinical care rather than randomized observation.^{[1][2]}

Dataset audit

The official overview makes clear that a truly exhaustive item-level variable list is dictionary-driven: MIMIC-III stores many measurement concepts through ITEMIDs and LABITEMIDs that must be resolved through D_ITEMS and D_LABITEMS, so the right audit granularity for causal design is table/domain plus clinically standard concept families extracted from those dictionaries. MIMIC-III contains 26 tables, with five stay-defining tables, five dictionary tables, and the remainder devoted to patient care events, diagnoses, procedures, medications, outputs, notes, and labs.^[1]

1. Dataset audit table

Table / domain	Variable type	Clinically relevant fields or families	Unit	Notes
PATIENTS ^[1]	Background / outcome	SUBJECT_ID, sex, date of birth / age proxy, date of death, patient-level expire flag	N/A	Patients older than 89 are age-shifted and can appear older than 300 years because of deidentification ^[1] .
ADMISSIONS ^[1]	Background / outcome / care-process	HADM_ID, admission/discharge time, admission type, ethnicity, diagnosis text, in-hospital death fields, ED times where available	N/A	Hospital mortality and admission context live here; out-of-hospital mortality is linked from external death data ^[1] .
ICUSTAYS ^[1]	Background / care-process	ICUSTAY_ID, ICU intime/outtime, first/last care unit, ICU LOS, source system	Time, days	Defines ICU episode and ICU type; important for baseline case-mix and measurement intensity ^[1] .
SERVICES ^[1]	Background / care-process	Service assignments and changes	N/A	Useful for medical vs surgical context and transitions of care ^[1] .
TRANSFERS ^[1]	Background / care-process	Ward / unit movement history	N/A	Useful for pre-ICU trajectory and unit exposure ^[1] .
CHAREVENTS ^[1]	Observed time-varying	Nurse-verified bedside physiology and charted observations, such as heart rate, arterial blood pressure, respiratory rate, and other bedside measures	Varies by ITEMID	The official description gives examples of hourly charting and notes roughly one bedside vital datapoint per hour on average ^[1] .
LABEVENTS + D_LABITEMS ^[1]	Observed time-varying	Chemistry, hematology, and other laboratory tests	VALUEUOM / item-specific	Lab concepts are resolved through D_LABITEMS; units vary by test ^[1] .
MICROBIOLOGYEVENTS ^[1]	Observed time-varying / infection process	Culture results, organism fields, specimen-level microbiology	Varies	Negative cultures are recorded with null organism name in v1.4 ^[1] .
INPUTEVENTS_CV / INPUTEVENTS_MV ^[1]	Treatment / care-process	Continuous IV drips, fluids, medication infusions, amounts and rates	Varies by drug/fluid	CareVue and MetaVision are stored separately for inputs because source structure differed ^[1] .
OUTPUTEVENTS ^[1]	Observed time-varying / care-process	Urine output and other outputs	Varies	This is central for renal and volume-status proxies ^[1] .

Table / domain	Variable type	Clinically relevant fields or families	Unit	Notes
PROCEDUREEVENTS_MV [1]	Treatment / care-process	Procedures with start/stop time, including ventilation, dialysis, x-rays, and others	N/A	Especially useful for respiratory support and renal replacement proxies in MetaVision patients [1] .
PRESCRIPTIONS [1]	Treatment / care-process	Medication orders	Varies	Useful for antibiotics and chronic medication context, though order timing is coarser than bedside events [1] .
NOTEEVENTS [1]	Treatment / care-process / unstructured observation	Discharge summaries, ECG reports, imaging reports, provider notes	Text	Valuable clinically, but high leakage risk for early-state causal tagging unless aggressively time-filtered [1] .
DIAGNOSES_ICD / PROCEDURES_ICD / DRGCODES / CPTEVENTS [1]	Background / treatment / billing-derived context	ICD-9 diagnoses, coded procedures, DRG, CPT	Codes	Best used for chronic burden and admission context, not as direct physiologic states [1] .
D_ITEMS / D_LABITEMS / D_ICD_* / D_CPT [1]	Dictionary	Concept definitions and code lookups	N/A	Required to harmonize ITEMIDs and derive clinically named variables [1] .

Compact audit summary

- Static/background variables clearly supported by the official docs: age proxy, sex, ethnicity, admission context, ICU stay metadata, service/transfer history, diagnoses and procedure codes. [\[1\]](#)
- Time-varying observed measurements clearly supported: bedside vitals and charted observations, laboratory tests, microbiology, outputs, and procedure times. [\[1\]](#)
- Outcomes clearly supported: in-hospital mortality, hospital discharge timing, and post-discharge mortality linkage through the Social Security Death Master File. [\[1\]](#)
- Treatments/care-process variables clearly supported: infusions, fluids, medication orders, procedures including ventilation and dialysis, and notes/reports. [\[1\]](#)
- Measurement-process concerns are intrinsic because MIMIC-III was populated from routine hospital care rather than a study protocol, and because bedside charting, labs, and procedures arise from clinician workflow and suspicion. [\[1\]](#)
- Many ML pipelines use reduced extracts rather than the full database; one benchmark subset contains over 31 million events and 17 selected clinical variables across 42,276 ICU stays, so the latent ontology should degrade gracefully to sparse extracts while still being anchored in the full database design. [\[3\]](#)

Latent ontology

The official database structure strongly supports latent states that sit between background risk and observed measurements, because MIMIC stores many noisy manifestations across vitals, labs, outputs, procedures, and coded context rather than a small set of directly observed disease states. I therefore recommend a core ontology centered on chronic vulnerability, inflammatory burden, organ-specific dysfunction, and a downstream global severity node. [\[1\]](#)

2. Candidate latent variables

Candidate latent variable	Clinical meaning	Supporting observed variables / metadata	Dataset support	Keep / merge / optional / reject	Rationale
LAT_CHRONIC_RISK	Baseline vulnerability before the acute ICU insult	Age, sex, admission history, ICD-9 chronic disease burden, pre-ICU trajectory	High	Keep	Strongly supported by demographics, admissions, ICU metadata, and diagnoses tables [1] .
LAT_ACUTE_INFLAMMATION_SEPSIS	Infection/inflammatory burden driving acute deterioration	Temperature, WBC-family labs, cultures, antibiotics, lactate, infection diagnoses	Medium-high	Keep	MIMIC explicitly contains labs, microbiology, prescriptions, and diagnoses, making this recoverable as a latent syndrome [1] .

Candidate latent variable	Clinical meaning	Supporting observed variables / metadata	Dataset support	Keep / merge / optional / reject	Rationale
LAT_SHOCK	Hemodynamic instability / impaired perfusion	BP/MAP, heart rate, lactate, urine output, vasopressor infusions	High	Keep	Bedside vitals, outputs, and infusion tables directly support a shock proxy 14 .
LAT_RESP_FAILURE	Pulmonary dysfunction and gas-exchange failure	RR, oxygen saturation, blood gases where available, ventilation procedures	High	Keep	Respiratory bedside measurements plus ventilation procedures are well supported 14 .
LAT_RENAL_DYSFUNCTION	Acute kidney injury / renal failure burden	Creatinine-family labs, BUN, potassium/bicarbonate adjuncts, urine output, dialysis	High	Keep	Labs, outputs, and dialysis procedures support a robust renal latent 14 .
LAT_NEURO_DYSFUNCTION	Impaired consciousness / neurologic failure	GCS-family charting, neurologic charted state	Medium	Keep	CHARTEVENTS contains bedside charting and the release notes specifically discuss GCS concepts, so this is usable though noisier than shock/renal/respiratory states 14 .
LAT_METABOLIC_DERANGEMENT	Acid-base, lactate, glucose, and electrolyte instability	pH, bicarbonate/total CO2, lactate, glucose, potassium, anion gap	Medium-high	Keep	Lab tables support a metabolic latent that mediates multiple organ failures 14 .
LAT_GLOBAL_SEVERITY	Multisystem physiologic instability	Cross-domain abnormalities and support intensity	High	Keep	Needed to summarize shared severity that explains broad abnormal measurement patterns and mortality 14 .
LAT_COAG_DYSFUNCTION	Hematologic/coagulopathic failure	Hematology/coag labs, bleeding-related procedures	Medium	Optional	Likely available through lab tables, but item harmonization is less universal across extracts 14 .
LAT_HEPATIC_DYSFUNCTION	Liver injury / synthetic dysfunction	Chemistry labs and coded liver disease context	Medium	Optional	Plausible in full MIMIC, but weaker in reduced benchmark-style extracts 14 15 .
LAT_CARDIAC_INJURY	Acute myocardial injury / strain	ECG reports, cardiac labs, vasoactive support	Low-medium	Reject as core node	Important clinically, but less reliably identifiable from standard structured extracts without careful concept engineering 14 .
LAT_VOLUME_STATUS	Hypovolemia / fluid overload	Inputs, outputs, weight, hemodynamics	Medium	Merge into shock/global severity	Real but difficult to separate from shock and treatment in routine ICU data.
LAT_CARE_INTENSITY	Monitoring and intervention intensity	Lab frequency, charting density, procedure density, ICU type	High	Do not model as biologic latent; use as measurement-process node	This is an observation-process variable, not a patient physiologic state 14 .

3. Final approved latent variables

Latent variable	Clinical meaning	Observed indicators	Suggested definition	Missingness strategy	Confidence	
LAT_CHRONIC_RISK	Pre-ICU vulnerability from age and chronic disease burden	Age, chronic ICD groups, pre-ICU LOS, emergency/urgent admission	Binary score over baseline risk domains	Missing baseline codes count as 0; keep separate QC flag	High	Be tre co ca

Latent variable	Clinical meaning	Observed indicators	Suggested definition	Missingness strategy	Confidence	
LAT_ACUTE_INFLAMMATION_SEPSIS	Acute inflammatory/infectious burden	Temp, WBC, lactate, cultures, antibiotics, infection ICDs	Binary syndrome proxy requiring infection evidence or strong inflammatory pattern	Missing culture/lab data handled conservatively	Medium-high	Ca up rat do org
LAT_SHOCK	Circulatory failure and tissue hypoperfusion	MAP/SBP, HR, lactate, urine output, vasopressor use	Binary score over hemodynamic/perfusion/support domains	Missing lactate or UO lowers sensitivity; do not infer from BP alone unless severe	High	Sti an su
LAT_RESP_FAILURE	Hypoxemic or ventilatory respiratory failure	SpO2, RR, PaO2/FiO2 or PaO2, PaCO2/pH, ventilation	Binary score over oxygenation, ventilatory work, support, gas exchange	Missing ABG tolerated if pulse ox and ventilation data exist	High	Sti ph dir ch da
LAT_RENAL_DYSFUNCTION	AKI or clinically important renal failure	Creatinine, BUN, urine output, potassium, bicarbonate, dialysis	Binary score over filtration, output, and renal-consequence domains	Missing UO tolerated if chemistry is available	High	Sti dy lat sh
LAT_NEURO_DYSFUNCTION	Depressed consciousness / neurologic impairment	GCS total and components	Binary proxy from GCS domains	If GCS unavailable, output 0 with low-evidence flag	Medium	Us se co lin ide
LAT_METABOLIC_DERANGEMENT	Acid-base and biochemical instability	pH, HCO3/CO2, lactate, glucose, anion gap, potassium	Binary score over acid-base, lactate, gap, glucose extremes	Missing ABG tolerated if chemistry supports label	Medium-high	Im me org mc
LAT_GLOBAL_SEVERITY	Multisystem instability across domains	Cross-domain physiologic abnormalities and support	Binary multisystem score	Requires at least partial coverage of several organ domains	High	Ol pri m: us ad co

Binary proxies

These trees are implementation devices for latent tags, not the scientific endpoint. Each tree uses only observed variables or metadata plausibly available in standard MIMIC concept extraction from CHARTEVENTS, LABEVENTS, INPUTEVENTS, OUTPUTEVENTS, PROCEDUREEVENTS, PRESCRIPTIONS, and coded tables.^[1]

4. Rule-based binary decision trees

LAT_CHRONIC_RISK

- Label 1: high baseline vulnerability before ICU physiology is considered.
- Inputs: age_at_icu, admission_type, preICU_LOS_hours, chronic ICD groups.
- Domain indicators:
 - D_1 : age_at_icu ≥ 75
 - D_2 : emergency_or_urgent_admission = 1
 - D_3 : preICU_LOS_hours > 24
 - D_4 : chronic_disease_group_count ≥ 2 , where groups may include CHF, COPD, CKD, cirrhosis, metastatic cancer, diabetes with complications
- Score: score = $D_1 + D_2 + D_3 + D_4$
- Threshold: LAT_CHRONIC_RISK = 1 if score ≥ 2 else 0
- Missingness policy: missing ICD groups count as 0; preserve QC_CHRONIC_CODES_MISSING flag.
- Pseudocode:

```
def lat_chronic_risk(x):
    d1 = x.age_at_icu >= 75
    d2 = x.admission_type in {"EMERGENCY", "URGENT"}
    d3 = x.preicu_los_hours is not None and x.preicu_los_hours > 24
```



```
d4 = x.chronic_disease_group_count >= 2
return int(sum([d1,d2,d3,d4]) >= 2)
```

```
flowchart TD
A[Age >= 75?] --> S[score = D1 + D2 + D3 + D4]
B[Emergency/Urgent?] --> S
C[Pre-ICU LOS > 24h?] --> S
D[>=2 chronic disease groups?] --> S
S --> T{score >= 2?}
T -->|Yes| Y[LAT_CHRONIC_RISK = 1]
T -->|No| N[LAT_CHRONIC_RISK = 0]
```

- Clinical rationale: pre-existing vulnerability modifies tolerance to acute insults.
- False positives: elective but frail chronic disease patients.
- False negatives: younger patients with severe unmeasured frailty.
- Confidence: high.

LAT_ACUTE_INFLAMMATION_SEPSIS

- Label 1: clinically meaningful inflammatory/infectious burden.
- Inputs: temp_min/max_24h, wbc_min/max_24h, lactate_max_24h, culture_ordered_or_positive_24h, antibiotics_any_24h, infection_icd_any.
- Domain indicators:
 - D_1 : infection_evidence = positive_or_ordered_culture OR antibiotics_any OR infection_icd_any
 - D_2 : temp_max > 38.3 OR temp_min < 36.0
 - D_3 : wbc_max > 12 OR wbc_min < 4
 - D_4 : lactate_max > 2
- Score: score = $D_1 + D_2 + D_3 + D_4$
- Threshold: LAT_ACUTE_INFLAMMATION_SEPSIS = 1 if (D_1 and score >= 2) OR score >= 3
- Missingness policy: if culture data absent, allow antibiotics or infection ICD as infection evidence; if all infection evidence missing, require 3 physiologic domains.
- Pseudocode:

```
def lat_inflammation_sepsis(x):
    d1 = x.culture_any or x.antibiotics_any or x.infection_icd_any
    d2 = (x.temp_max is not None and x.temp_max > 38.3) or (x.temp_min is not None and x.temp_min < 36.0)
    d3 = (x.wbc_max is not None and x.wbc_max > 12) or (x.wbc_min is not None and x.wbc_min < 4)
    d4 = x.lactate_max is not None and x.lactate_max > 2
    score = sum([d1,d2,d3,d4])
    return int((d1 and score >= 2) or (score >= 3))
```

```
flowchart TD
A[Infection evidence?] --> S[score = D1 + D2 + D3 + D4]
B[Temp abnormal?] --> S
C[WBC abnormal?] --> S
D[Lactate > 2?] --> S
S --> T{(D1 and score >= 2) or score >= 3?}
T -->|Yes| Y[LAT_ACUTE_INFLAMMATION_SEPSIS = 1]
T -->|No| N[LAT_ACUTE_INFLAMMATION_SEPSIS = 0]
```

- Clinical rationale: separates upstream inflammatory burden from downstream organ failures.
- False positives: sterile inflammation, postoperative stress.
- False negatives: culture-negative sepsis with sparse labs.
- Confidence: medium-high.

LAT_SHOCK

- Label 1: circulatory shock or clinically significant hemodynamic instability.
- Inputs: map_min_6h, sbp_min_6h, hr_max_6h, lactate_max_24h, urine_output_mkgph_6h, vasopressor_any_24h.
- Domain indicators:
 - D_1 : map_min < 65 OR sbp_min < 90
 - D_2 : hr_max > 120

- D_3 : lactate_max > 2
- D_4 : urine_output_mlgph_6h < 0.5
- D_5 : vasopressor_any = 1
- Score: score = $D_1 + D_2 + D_3 + D_4 + D_5$
- Threshold: LAT_SHOCK = 1 if score >= 2 and D_1 is true, or if D_5 is true
- Missingness policy: missing lactate/UO does not block positivity if hypotension and vasopressor support are present.
- Pseudocode:

```
def lat_shock(x):
    d1 = (x.map_min is not None and x.map_min < 65) or (x.sbp_min is not None and x.sbp_min < 90)
    d2 = x.hr_max is not None and x.hr_max > 120
    d3 = x.lactate_max is not None and x.lactate_max > 2
    d4 = x.uo_mlgph_6h is not None and x.uo_mlgph_6h < 0.5
    d5 = x.vasopressor_any
    score = sum([d1,d2,d3,d4,d5])
    return int((d1 and score >= 2) or d5)
```

```
flowchart TD
    A[MAP < 65 or SBP < 90?] --> S[score = D1 + D2 + D3 + D4 + D5]
    B[HR > 120?] --> S
    C[Lactate > 2?] --> S
    D[Urine output low?] --> S
    E[Vasopressor used?] --> S
    S --> T1{(D1 and score >= 2) or D5?}
    T1 -->|Yes| Y[LAT_SHOCK = 1]
    T1 -->|No| N[LAT_SHOCK = 0]
```

- Clinical rationale: blends hypotension, hypoperfusion, renal hypoperfusion, and treatment support.
- False positives: transient peri-intubation hypotension, chronic low BP.
- False negatives: compensated shock without documented lactate or UO.
- Confidence: high.

LAT_RESP_FAILURE

- Label 1: clinically significant hypoxemic or ventilatory respiratory failure.
- Inputs: spo2_min_6h, rr_min/max_6h, pao2_min_24h, fio2_max_24h, pfratio_min_24h, paco2_max_24h, ph_min_24h, mechvent_any_24h.
- Domain indicators:
 - D_1 : spo2_min < 90 OR pao2_min < 60 OR pfratio_min < 300
 - D_2 : rr_max > 30 OR rr_min < 8
 - D_3 : mechvent_any = 1
 - D_4 : paco2_max > 50 AND ph_min < 7.30
- Score: score = $D_1 + D_2 + D_3 + D_4$
- Threshold: LAT_RESP_FAILURE = 1 if score >= 2
- Missingness policy: ABG-derived domains can be missing; pulse ox + RR + ventilation can still activate the label.
- Pseudocode:

```
def lat_resp_failure(x):
    d1 = (x.spo2_min is not None and x.spo2_min < 90) or \
        (x.pao2_min is not None and x.pao2_min < 60) or \
        (x.pfratio_min is not None and x.pfratio_min < 300)
    d2 = (x.rr_max is not None and x.rr_max > 30) or (x.rr_min is not None and x.rr_min < 8)
    d3 = x.mechvent_any
    d4 = (x.paco2_max is not None and x.paco2_max > 50) and (x.ph_min is not None and x.ph_min < 7.30)
    return int(sum([d1,d2,d3,d4]) >= 2)
```

```
flowchart TD
    A[Oxygenation abnormal?] --> S[score = D1 + D2 + D3 + D4]
    B[RR abnormal?] --> S
    C[Mechanical ventilation?] --> S
    D[Hypercapnic acidosis?] --> S
    S --> T{score >= 2?}
```

```
T -->|Yes| Y[LAT_RESP_FAILURE = 1]
T -->|No| N[LAT_RESP_FAILURE = 0]
```

- Clinical rationale: captures both gas-exchange failure and ventilatory pump failure.
- False positives: transient desaturation or prophylactic ventilation.
- False negatives: occult respiratory fatigue before support.
- Confidence: high.

LAT_RENAL_DYSFUNCTION

- Label 1: clinically important renal dysfunction.
- Inputs: creat_max_24h, creat_delta_48h, bun_max_24h, uo_mlgph_6h, potassium_max_24h, hco3_min_24h, dialysis_any_24h.
- Domain indicators:
 - D_1 : creat_max $\geq$ 2.0 OR creat_delta_48h $\geq$ 0.3
 - D_2 : bun_max $\geq$ 40
 - D_3 : uo_mlgph_6h $<$ 0.5
 - D_4 : potassium_max $\geq$ 5.5 OR hco3_min $<$ 18
 - D_5 : dialysis_any = 1
- Score: score = $D_1 + D_2 + D_3 + D_4 + D_5$
- Threshold: LAT_RENAL_DYSFUNCTION = 1 if score $\geq$ 2 OR D_5
- Missingness policy: if UO missing, rely on chemistry and dialysis; if no chemistry but dialysis present, positive.
- Pseudocode:

```
def lat_renal_dysfunction(x):
    d1 = (x.creat_max is not None and x.creat_max >= 2.0) or \
        (x.creat_delta_48h is not None and x.creat_delta_48h >= 0.3)
    d2 = x.bun_max is not None and x.bun_max >= 40
    d3 = x.uo_mlgph_6h is not None and x.uo_mlgph_6h < 0.5
    d4 = (x.k_max is not None and x.k_max >= 5.5) or (x.hco3_min is not None and x.hco3_min < 18)
    d5 = x.dialysis_any
    return int(sum([d1,d2,d3,d4,d5]) >= 2 or d5)
```

```
flowchart TD
    A[Creatinine abnormal?] --> S[score = D1 + D2 + D3 + D4 + D5]
    B[BUN high?] --> S
    C[Urine output low?] --> S
    D[Renal consequence labs abnormal?] --> S
    E[Dialysis used?] --> S
    S --> T{score >= 2 or D5?}
    T -->|Yes| Y[LAT_RENAL_DYSFUNCTION = 1]
    T -->|No| N[LAT_RENAL_DYSFUNCTION = 0]
```

- Clinical rationale: filtration failure plus output failure plus physiologic consequences.
- False positives: chronic kidney disease without acute worsening.
- False negatives: early AKI before creatinine rise.
- Confidence: high.

LAT_NEURO_DYSFUNCTION

- Label 1: clinically meaningful depressed consciousness or neurologic dysfunction.
- Inputs: gcs_min_6h, gcs_eye_min, gcs_verbal_min, gcs_motor_min.
- Domain indicators:
 - D_1 : gcs_min $<$ 13
 - D_2 : gcs_min $\leq$ 8
 - D_3 : gcs_eye_min $\leq$ 2 OR gcs_verbal_min $\leq$ 3 OR gcs_motor_min $\leq$ 5
- Score: score = $D_1 + D_2 + D_3$
- Threshold: LAT_NEURO_DYSFUNCTION = 1 if score $\geq$ 2
- Missingness policy: if all GCS fields missing, output 0 and set QC_GCS_MISSING = 1.
- Pseudocode:

```
def lat_neuro_dysfunction(x):
    d1 = x.gcs_min is not None and x.gcs_min < 13
    d2 = x.gcs_min is not None and x.gcs_min <= 8
    d3 = (x.gcs_eye_min is not None and x.gcs_eye_min <= 2) or \
        (x.gcs_verbal_min is not None and x.gcs_verbal_min <= 3) or \
        (x.gcs_motor_min is not None and x.gcs_motor_min <= 5)
    return int(sum([d1,d2,d3]) >= 2)
```

```
flowchart TD
    A[GCS < 13?] --> S[score = D1 + D2 + D3]
    B[GCS <= 8?] --> S
    C[Severe GCS component abnormality?] --> S
    S --> T[score >= 2?]
    T -->|Yes| Y[LAT_NEURO_DYSFUNCTION = 1]
    T -->|No| N[LAT_NEURO_DYSFUNCTION = 0]
```

- Clinical rationale: uses bedside neurologic charting without notes.
- False positives: sedation, intubation-related verbal suppression.
- False negatives: focal neurologic deficits with preserved GCS.
- Confidence: medium.

LAT_METABOLIC_DERANGEMENT

- Label 1: clinically important acid-base or biochemical instability.
- Inputs: ph_min_24h, hco3_min_24h, lactate_max_24h, anion_gap_max_24h, glucose_min/max_24h, potassium_max_24h.
- Domain indicators:
 - D_1 : ph_min < 7.30 OR hco3_min < 18
 - D_2 : lactate_max > 2
 - D_3 : anion_gap_max > 16
 - D_4 : glucose_min < 60 OR glucose_max > 250
 - D_5 : potassium_max >= 5.5
- Score: score = $D_1 + D_2 + D_3 + D_4 + D_5$
- Threshold: LAT_METABOLIC_DERANGEMENT = 1 if score >= 2
- Missingness policy: if ABG missing, chemistry-only rule still allowed.
- Pseudocode:

```
def lat_metabolic_derangement(x):
    d1 = (x.ph_min is not None and x.ph_min < 7.30) or (x.hco3_min is not None and x.hco3_min < 18)
    d2 = x.lactate_max is not None and x.lactate_max > 2
    d3 = x.anion_gap_max is not None and x.anion_gap_max > 16
    d4 = (x.glucose_min is not None and x.glucose_min < 60) or (x.glucose_max is not None and x.glucose_max > 250)
    d5 = x.k_max is not None and x.k_max >= 5.5
    return int(sum([d1,d2,d3,d4,d5]) >= 2)
```

```
flowchart TD
    A[Acid-base abnormal?] --> S[score = D1 + D2 + D3 + D4 + D5]
    B[Lactate > 2?] --> S
    C[Anion gap high?] --> S
    D[Glucose extreme?] --> S
    E[Potassium high?] --> S
    S --> T[score >= 2?]
    T -->|Yes| Y[LAT_METABOLIC_DERANGEMENT = 1]
    T -->|No| N[LAT_METABOLIC_DERANGEMENT = 0]
```

- Clinical rationale: downstream integrator of shock, respiratory failure, and renal dysfunction.
- False positives: chronic metabolic disorders.
- False negatives: compensated mixed disorders.
- Confidence: medium-high.

LAT_GLOBAL_SEVERITY

- Label 1: multisystem instability.
- Inputs: domain-level observed summaries, not mortality and not future outcomes.
- Domain indicators:
 - D_1 : circulatory domain positive, defined as $\text{MAP} < 65$ OR vasopressor_any
 - D_2 : respiratory domain positive, defined as $\text{SpO}_2 < 90$ OR mechvent_any OR $\text{PF} < 300$
 - D_3 : renal domain positive, defined as $\text{creatinine} \geq 2$ OR $\text{UO} < 0.5$
 - D_4 : neurologic domain positive, defined as $\text{GCS} < 13$
 - D_5 : metabolic/inflammatory domain positive, defined as $\text{lactate} > 2$ OR $\text{pH} < 7.30$ OR $\text{HCO}_3 < 18$ OR severe temperature/WBC abnormality
- Score: $\text{score} = D_1 + D_2 + D_3 + D_4 + D_5$
- Threshold: $\text{LAT_GLOBAL_SEVERITY} = 1$ if $\text{score} \geq 3$
- Missingness policy: require at least 3 observed domains; otherwise output 0 with $\text{QC_GLOBAL_LOW_COVERAGE} = 1$.
- Pseudocode:

```
def lat_global_severity(x):
    d1 = ((x.map_min is not None and x.map_min < 65) or x.vasopressor_any)
    d2 = ((x.spo2_min is not None and x.spo2_min < 90) or x.mechvent_any or (x.pfratio_min is not None
    d3 = ((x.creat_max is not None and x.creat_max >= 2.0) or (x.uo_mlgph_6h is not None and x.uo_mlgph_6h < 0.5))
    d4 = (x.gcs_min is not None and x.gcs_min < 13)
    d5 = ((x.lactate_max is not None and x.lactate_max > 2) or
          (x.ph_min is not None and x.ph_min < 7.30) or
          (x.hco3_min is not None and x.hco3_min < 18) or
          ((x.temp_max is not None and x.temp_max > 38.3) or (x.temp_min is not None and x.temp_min < 36.0) or
          (x.wbc_max is not None and x.wbc_max > 12) or (x.wbc_min is not None and x.wbc_min < 4)))
    observed = sum(v is not None for v in [x.map_min, x.spo2_min, x.creat_max, x.gcs_min, x.lactate_max, x.ph_min, x.hco3_min, x.temp_max, x.temp_min, x.wbc_max, x.wbc_min])
    score = sum([d1,d2,d3,d4,d5])
    return int(score >= 3 and observed >= 3)
```

```
flowchart TD
    A[Circulatory domain?] --> S[score = D1 + D2 + D3 + D4 + D5]
    B[Respiratory domain?] --> S
    C[Renal domain?] --> S
    D[Neurologic domain?] --> S
    E[Metabolic/Inflammatory domain?] --> S
    S --> T{score >= 3?}
    T -->|Yes| Y[LAT_GLOBAL_SEVERITY = 1]
    T -->|No| N[LAT_GLOBAL_SEVERITY = 0]
```

- Clinical rationale: proxy for shared instability across organ systems.
- False positives: transient multi-domain abnormalities around procedures.
- False negatives: severe illness with sparse observation.
- Confidence: high.

Measurement and DAG

MIMIC-III's structure supports a measurement model in which latent clinical states generate multiple observed proxies across charts, labs, outputs, procedures, and medications, while background variables and care context also influence what gets measured and how often. Because the database is derived from routine clinical care and includes ICU source-system differences, care-unit context, bedside charting, and procedure documentation, missingness and measurement intensity should be modeled as informative descendants of severity and clinician decision-making rather than ignored noise.^[1]

5. Observed-to-latent measurement model

Observed variable	Parent latent(s)	Direction of abnormality	Specific?	Used in binary tree?	Notes
MAP / SBP	LAT_SHOCK, LAT_GLOBAL_SEVERITY	Lower worse	Moderately specific	Yes	Hypotension is core circulatory evidence.
Heart rate	LAT_SHOCK, LAT_ACUTE_INFLAMMATION_SEPSIS, LAT_GLOBAL_SEVERITY	Higher worse	Non-specific	Yes	Tachycardia is shared stress response.

Observed variable	Parent latent(s)	Direction of abnormality	Specific?	Used in binary tree?	Notes
Lactate	LAT_SHOCK, LAT_ACUTE_INFLAMMATION_SEPSIS, LAT_METABOLIC_DERANGEMENT, LAT_GLOBAL_SEVERITY	Higher worse	Non-specific	Yes	Multi-parent proxy by design.
Urine output	LAT_SHOCK, LAT_RENAL_DYSFUNCTION, LAT_GLOBAL_SEVERITY	Lower worse	Moderately specific	Yes	Could reflect renal hypoperfusion or intrinsic renal failure.
Vasopressor use	LAT_SHOCK, MISS_MEAS_INTENSITY	Presence worse / more severe	Non-specific	Yes	Treatment proxy; use only if exposure of interest is not vasopressors.
SpO2	LAT_RESP_FAILURE, LAT_GLOBAL_SEVERITY	Lower worse	Moderately specific	Yes	



Complete the DAG construction for MIMIC-III dataset. Showcase your results in details - divided to headers:

- DAG, mortality paths, and causal implications
- Core edge list
- Mortality mechanism
- Adjacency list
- Mermaid DAG

Below is a clinically plausible latent-variable DAG for MIMIC-III that treats ICU measurements as manifestations of hidden physiologic states and treats mortality as downstream of background risk, acute insult, organ dysfunction, and care-process intensity rather than as a direct function of raw labs alone. Because MIMIC-III is a routine-care ICU database with bedside charting, labs, inputs, outputs, procedures, coded diagnoses, and mortality fields, the graph should explicitly include observation-process nodes for measurement intensity and missingness, which are partly driven by illness severity and ICU workflow.^[28]

DAG, mortality paths, and causal implications

The recommended causal backbone is: background vulnerability and admission context → acute inflammatory burden and latent global severity → organ-specific dysfunction states → mortality, with organ states simultaneously generating the observed labs, vitals, outputs, and support variables captured in MIMIC-III. This keeps the graph sparse, clinically interpretable, and aligned with the fact that MIMIC-III stores many event tables that reflect routine care processes rather than direct biologic truth.^[28]

Key causal implications follow from that structure: LAT_CHRONIC_RISK is usually a pre-exposure confounder; LAT_GLOBAL_SEVERITY and organ-failure nodes are often mediators or outcome-proximal severity markers; and MISS_MEAS_INTENSITY is commonly a collider or descendant of both severity and care setting, so conditioning on it can induce bias when estimating treatment or physiology effects on mortality. In practical causal analyses, adjustment should usually favor background variables and pre-ICU burden over downstream ICU physiologic tags unless the estimand explicitly requires controlled direct effects through mediators.^[28]

Core edge list

The edge list below is the core latent DAG, not the full item-level measurement graph. Each edge is included only when there is a clinically plausible temporal and mechanistic direction consistent with ICU data generation in MIMIC-III.^[28]

Source node	Target node	Edge type	Clinical mechanism	Confidence	Keep / optional
BG_AGE	LAT_CHRONIC_RISK	background-to-latent	Older age increases frailty, multimorbidity burden, and reduced physiologic reserve ^[28]	High	Keep

Source node	Target node	Edge type	Clinical mechanism	Confidence	Keep / optional
BG_SEX	LAT_CHRONIC_RISK	background-to-latent	Sex influences chronic disease distribution and baseline risk profiles [28]	Medium	Keep
BG_ETHNICITY	LAT_CHRONIC_RISK	background-to-latent	Captures structural case-mix and chronic disease heterogeneity in admissions data [28]	Medium	Keep
BG_ADMISSION_TYPE	LAT_ACUTE_INFLAMMATION_SEPSIS	background-to-latent	Emergency/urgent admissions are more likely to arrive with acute infection or systemic inflammatory burden [28]	Medium	Keep
BG_ICU_TYPE	MISS_MEAS_INTENSITY	measurement-process	ICU type changes monitoring density, test ordering, and support use [28]	High	Keep
BG_ICU_TYPE	LAT_GLOBAL_SEVERITY	background-to-latent	Different ICU populations have different expected severity and organ-failure mix [28]	Medium	Keep
BG_PREICU_TRAJECTORY	LAT_CHRONIC_RISK	background-to-latent	Longer pre-ICU hospital course suggests baseline vulnerability or deterioration before ICU [28]	Medium	Keep
LAT_CHRONIC_RISK	LAT_GLOBAL_SEVERITY	latent-to-latent	Lower reserve increases severity for a given acute insult [28]	High	Keep
LAT_CHRONIC_RISK	LAT_RENAL_DYSFUNCTION	latent-to-latent	CKD/frailty predispose to renal decompensation during acute illness [28]	Medium-high	Keep
LAT_CHRONIC_RISK	LAT_RESP_FAILURE	latent-to-latent	Baseline cardiopulmonary burden increases risk of respiratory failure [28]	Medium	Keep
LAT_CHRONIC_RISK	OUT_MORTALITY	latent-to-mortality	Frailty and chronic burden raise mortality even beyond acute physiology [28]	High	Keep
LAT_ACUTE_INFLAMMATION_SEPSIS	LAT_GLOBAL_SEVERITY	latent-to-latent	Systemic inflammation drives broad physiologic instability [28]	High	Keep
LAT_ACUTE_INFLAMMATION_SEPSIS	LAT_SHOCK	latent-to-latent	Sepsis causes vasodilation, capillary leak, and distributive shock [28]	High	Keep
LAT_ACUTE_INFLAMMATION_SEPSIS	LAT_RESP_FAILURE	latent-to-latent	Pneumonia/ARDS-like inflammatory injury impairs oxygenation [28]	Medium-high	Keep

Source node	Target node	Edge type	Clinical mechanism	Confidence	Keep / optional
LAT_ACUTE_INFLAMMATION_SEPSIS	LAT_RENAL_DYSFUNCTION	latent-to-latent	Sepsis promotes AKI through hypoperfusion, inflammation, and tubular injury [28]	Medium-high	Keep
LAT_ACUTE_INFLAMMATION_SEPSIS	LAT_NEURO_DYSFUNCTION	latent-to-latent	Sepsis-associated encephalopathy contributes to depressed consciousness [28]	Medium	Keep
LAT_ACUTE_INFLAMMATION_SEPSIS	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Infection and tissue hypoperfusion elevate lactate and acid-base stress [28]	High	Keep
LAT_GLOBAL_SEVERITY	LAT_SHOCK	latent-to-latent	Shared multisystem instability increases hemodynamic failure risk [28]	High	Keep
LAT_GLOBAL_SEVERITY	LAT_RESP_FAILURE	latent-to-latent	Global severity often includes pulmonary compromise and need for support [28]	High	Keep
LAT_GLOBAL_SEVERITY	LAT_RENAL_DYSFUNCTION	latent-to-latent	Severe critical illness precipitates kidney injury through low reserve and hypoperfusion [28]	High	Keep
LAT_GLOBAL_SEVERITY	LAT_NEURO_DYSFUNCTION	latent-to-latent	Severe illness contributes to encephalopathy and depressed neurologic status [28]	Medium-high	Keep
LAT_GLOBAL_SEVERITY	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Severe illness disrupts acid-base, glucose, and biochemical homeostasis [28]	High	Keep
LAT_SHOCK	LAT_RENAL_DYSFUNCTION	latent-to-latent	Renal hypoperfusion drives AKI and oliguria [28]	High	Keep
LAT_SHOCK	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Tissue hypoperfusion elevates lactate and metabolic acidosis [28]	High	Keep
LAT_SHOCK	LAT_NEURO_DYSFUNCTION	latent-to-latent	Cerebral hypoperfusion worsens mental status in severe shock [28]	Medium	Keep
LAT_RESP_FAILURE	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypoxemia or hypercapnia contributes to respiratory and mixed acidosis [28]	High	Keep

Source node	Target node	Edge type	Clinical mechanism	Confidence	Keep / optional
LAT_RENAL_DYSFUNCTION	LAT_METABOLIC_DERANGEMENT	latent-to-latent	AKI causes acidosis, hyperkalemia, and biochemical instability ^[28]	High	Keep
LAT_SHOCK	OUT_MORTALITY	latent-to-mortality	Persistent circulatory failure directly raises risk of death ^[28]	High	Keep
LAT_RESP_FAILURE	OUT_MORTALITY	latent-to-mortality	Severe hypoxemia or ventilatory failure raises mortality ^[28]	High	Keep
LAT_RENAL_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	AKI and renal failure increase ICU and hospital mortality ^[28]	High	Keep
LAT_NEURO_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	Severe encephalopathy or coma is a strong mortality signal ^[28]	Medium-high	Keep
LAT_METABOLIC_DERANGEMENT	OUT_MORTALITY	latent-to-mortality	Severe acidosis, lactatemia, and electrolyte derangement contribute to death ^[28]	High	Keep
LAT_GLOBAL_SEVERITY	OUT_MORTALITY	latent-to-mortality	Multisystem physiologic instability is outcome-proximal ^[28]	High	Keep
LAT_CHRONIC_RISK	MISS_MEAS_INTENSITY	measurement-process	Frailer patients often receive denser monitoring or prolonged ICU observation ^[28]	Medium	Keep
LAT_GLOBAL_SEVERITY	MISS_MEAS_INTENSITY	measurement-process	Sicker patients generate more labs, charting, and procedures ^[28]	High	Keep
LAT_SHOCK	MISS_MEAS_INTENSITY	measurement-process	Shock triggers more hemodynamic checks, lactates, outputs, and infusions ^[28]	High	Keep
LAT_RESP_FAILURE	MISS_MEAS_INTENSITY	measurement-process	Respiratory instability increases gas measurements, pulse oximetry attention, and ventilation documentation ^[28]	High	Keep
LAT_RENAL_DYSFUNCTION	MISS_MEAS_INTENSITY	measurement-process	AKI increases chemistry, urine-output, and dialysis-related observation ^[28]	High	Keep

Mortality mechanism

The main mortality path in MIMIC-III should be read as a small number of clinically coherent pathways rather than many direct lab-to-death arrows. Background vulnerability increases susceptibility, acute inflammatory burden and global severity initiate decompensation, organ-specific failure states propagate physiologic collapse, and mortality follows when compensation fails across one or more critical systems. ^[28]

Important mortality paths in the final DAG are:

- BG_AGE → LAT_CHRONIC_RISK → LAT_GLOBAL_SEVERITY → OUT_MORTALITY, representing reduced reserve and frailty-driven vulnerability to critical illness.^[28]
- BG_ADMISSION_TYPE → LAT_ACUTE_INFLAMMATION_SEPSIS → LAT_SHOCK → LAT_RENAL_DYSFUNCTION → LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY, representing a septic shock to multiorgan-failure pathway.^[28]
- LAT_ACUTE_INFLAMMATION_SEPSIS → LAT_RESP_FAILURE → LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY, representing pulmonary inflammatory failure with downstream acidosis and biochemical collapse.^[28]
- LAT_GLOBAL_SEVERITY → LAT_NEURO_DYSFUNCTION → OUT_MORTALITY, representing encephalopathy or depressed consciousness as a downstream severity manifestation with prognostic relevance.^[28]
- LAT_GLOBAL_SEVERITY → LAT_SHOCK / LAT_RESP_FAILURE / LAT_RENAL_DYSFUNCTION → OUT_MORTALITY, representing parallel organ-failure channels from shared critical-illness burden.^[28]

Causally, this means most bedside variables in MIMIC-III are not clean causes of death on their own; they are noisy children of the latent states above, or descendants of treatment and monitoring processes captured in event tables. It also means LAT_GLOBAL_SEVERITY is often too downstream to use as a routine adjustment set for treatment effect estimation, because it will frequently mediate part of the effect from upstream exposures such as infection burden, oxygen therapy, vasopressors, or ventilation strategy.^[29] ^[28]

Adjacency list

This adjacency list gives the core latent DAG plus the measurement-process node. I keep observed measurement nodes out of the main adjacency list to preserve readability, but they should be attached downstream of the relevant latent parents as manifestations.^[28]

- BG_AGE: LAT_CHRONIC_RISK^[28]
- BG_SEX: LAT_CHRONIC_RISK^[28]
- BG_ETHNICITY: LAT_CHRONIC_RISK^[28]
- BG_ADMISSION_TYPE: LAT_ACUTE_INFLAMMATION_SEPSIS^[28]
- BG_ICU_TYPE: LAT_GLOBAL_SEVERITY, MISS_MEAS_INTENSITY^[28]
- BG_PREICU_TRAJECTORY: LAT_CHRONIC_RISK^[28]
- LAT_CHRONIC_RISK: LAT_GLOBAL_SEVERITY, LAT_RENAL_DYSFUNCTION, LAT_RESP_FAILURE, OUT_MORTALITY, MISS_MEAS_INTENSITY^[28]
- LAT_ACUTE_INFLAMMATION_SEPSIS: LAT_GLOBAL_SEVERITY, LAT_SHOCK, LAT_RESP_FAILURE, LAT_RENAL_DYSFUNCTION, LAT_NEURO_DYSFUNCTION, LAT_METABOLIC_DERANGEMENT^[28]
- LAT_GLOBAL_SEVERITY: LAT_SHOCK, LAT_RESP_FAILURE, LAT_RENAL_DYSFUNCTION, LAT_NEURO_DYSFUNCTION, LAT_METABOLIC_DERANGEMENT, OUT_MORTALITY, MISS_MEAS_INTENSITY^[28]
- LAT_SHOCK: LAT_RENAL_DYSFUNCTION, LAT_METABOLIC_DERANGEMENT, LAT_NEURO_DYSFUNCTION, OUT_MORTALITY, MISS_MEAS_INTENSITY^[28]
- LAT_RESP_FAILURE: LAT_METABOLIC_DERANGEMENT, OUT_MORTALITY, MISS_MEAS_INTENSITY^[28]
- LAT_RENAL_DYSFUNCTION: LAT_METABOLIC_DERANGEMENT, OUT_MORTALITY, MISS_MEAS_INTENSITY^[28]
- LAT_NEURO_DYSFUNCTION: OUT_MORTALITY^[28]
- LAT_METABOLIC_DERANGEMENT: OUT_MORTALITY^[28]

For the downstream measurement layer, the key pattern is: LAT_SHOCK → OBS_MAP, OBS_SBP, OBS_HR, OBS_LACTATE, OBS_URINE_OUTPUT, OBS_VASOPRESSOR_USE; LAT_RESP_FAILURE → OBS_SPO2, OBS_RR, OBS_PAO2, OBS_PF_RATIO, OBS_PACO2, OBS_MECHVENT; LAT_RENAL_DYSFUNCTION → OBS_CREATININE, OBS_BUN, OBS_URINE_OUTPUT, OBS_POTASSIUM, OBS_DIALYSIS; LAT_NEURO_DYSFUNCTION → OBS_GCS; LAT_METABOLIC_DERANGEMENT → OBS_PH, OBS_HCO3, OBS_ANION_GAP, OBS_GLUCOSE, OBS_LACTATE; and LAT_ACUTE_INFLAMMATION_SEPSIS → OBS_TEMP, OBS_WBC, OBS_CULTURES, OBS_ANTIBIOTICS, OBS_LACTATE.^[28]

Mermaid DAG

The diagram below is the compact causal graph I would use as the primary thesis DAG for MIMIC-III. It is intentionally sparse and keeps observed measurements as descendants of latent states rather than direct interacting causes.^[28]

```
graph TD
  BG_AGE --> LAT_CHRONIC_RISK
  BG_SEX --> LAT_CHRONIC_RISK
```

BG_ETHNICITY --> LAT_CHRONIC_RISK
BG_PREICU_TRAJECTORY --> LAT_CHRONIC_RISK

BG_ADMISSION_TYPE --> LAT_ACUTE_INFLAMMATION_SEPSIS
BG_ICU_TYPE --> LAT_GLOBAL_SEVERITY
BG_ICU_TYPE --> MISS_MEAS_INTENSITY

LAT_CHRONIC_RISK --> LAT_GLOBAL_SEVERITY
LAT_CHRONIC_RISK --> LAT_RENAL_DYSFUNCTION
LAT_CHRONIC_RISK --> LAT_RESP_FAILURE
LAT_CHRONIC_RISK --> OUT_MORTALITY
LAT_CHRONIC_RISK --> MISS_MEAS_INTENSITY

LAT_ACUTE_INFLAMMATION_SEPSIS --> LAT_GLOBAL_SEVERITY
LAT_ACUTE_INFLAMMATION_SEPSIS --> LAT_SHOCK
LAT_ACUTE_INFLAMMATION_SEPSIS --> LAT_RESP_FAILURE
LAT_ACUTE_INFLAMMATION_SEPSIS --> LAT_RENAL_DYSFUNCTION
LAT_ACUTE_INFLAMMATION_SEPSIS --> LAT_NEURO_DYSFUNCTION
LAT_ACUTE_INFLAMMATION_SEPSIS --> LAT_METABOLIC_DERANGEMENT

LAT_GLOBAL_SEVERITY --> LAT_SHOCK
LAT_GLOBAL_SEVERITY --> LAT_RESP_FAILURE
LAT_GLOBAL_SEVERITY --> LAT_RENAL_DYSFUNCTION
LAT_GLOBAL_SEVERITY --> LAT_NEURO_DYSFUNCTION
LAT_GLOBAL_SEVERITY --> LAT_METABOLIC_DERANGEMENT
LAT_GLOBAL_SEVERITY --> OUT_MORTALITY
LAT_GLOBAL_SEVERITY --> MISS_MEAS_INTENSITY

LAT_SHOCK --> LAT_RENAL_DYSFUNCTION
LAT_SHOCK --> LAT_METABOLIC_DERANGEMENT
LAT_SHOCK --> LAT_NEURO_DYSFUNCTION
LAT_SHOCK --> OUT_MORTALITY
LAT_SHOCK --> MISS_MEAS_INTENSITY

LAT_RESP_FAILURE --> LAT_METABOLIC_DERANGEMENT
LAT_RESP_FAILURE --> OUT_MORTALITY
LAT_RESP_FAILURE --> MISS_MEAS_INTENSITY

LAT_RENAL_DYSFUNCTION --> LAT_METABOLIC_DERANGEMENT
LAT_RENAL_DYSFUNCTION --> OUT_MORTALITY
LAT_RENAL_DYSFUNCTION --> MISS_MEAS_INTENSITY

LAT_NEURO_DYSFUNCTION --> OUT_MORTALITY
LAT_METABOLIC_DERANGEMENT --> OUT_MORTALITY

LAT_SHOCK --> OBS_MAP
LAT_SHOCK --> OBS_SBP
LAT_SHOCK --> OBS_HR
LAT_SHOCK --> OBS_LACTATE
LAT_SHOCK --> OBS_URINE_OUTPUT

LAT_RESP_FAILURE --> OBS_SPO2
LAT_RESP_FAILURE --> OBS_RR
LAT_RESP_FAILURE --> OBS_PAO2
LAT_RESP_FAILURE --> OBS_PF_RATIO
LAT_RESP_FAILURE --> OBS_MECHVENT

LAT_RENAL_DYSFUNCTION --> OBS_CREATININE
LAT_RENAL_DYSFUNCTION --> OBS_BUN
LAT_RENAL_DYSFUNCTION --> OBS_POTASSIUM
LAT_RENAL_DYSFUNCTION --> OBS_DIALYSIS

LAT_NEURO_DYSFUNCTION --> OBS_GCS

LAT_METABOLIC_DERANGEMENT --> OBS_PH
LAT_METABOLIC_DERANGEMENT --> OBS_HCO3
LAT_METABOLIC_DERANGEMENT --> OBS_ANION_GAP
LAT_METABOLIC_DERANGEMENT --> OBS_GLUCOSE

LAT_ACUTE_INFLAMMATION_SEPSIS --> OBS_TEMP
LAT_ACUTE_INFLAMMATION_SEPSIS --> OBS_WBC
LAT_ACUTE_INFLAMMATION_SEPSIS --> OBS_CULTURE
LAT_ACUTE_INFLAMMATION_SEPSIS --> OBS_ANTIBIOTICS

MISS_MEAS_INTENSITY --> OBS_MAP
MISS_MEAS_INTENSITY --> OBS_LACTATE
MISS_MEAS_INTENSITY --> OBS_PAO2
MISS_MEAS_INTENSITY --> OBS_CREATININE
MISS_MEAS_INTENSITY --> OBS_GCS

A useful reading rule for this DAG is: observed variables are mostly children of latent physiology and the observation process, not interchangeable causes of one another. That distinction is what makes the graph appropriate for causal inference rather than just predictive feature engineering in MIMIC-III. [29] [28]

✱

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